Single-Event Multinomial Full Kelly via Implicit State Positions

Abstract

For a single event with finitely many mutually exclusive outcomes, the full Kelly problem is to maximize expected log wealth over nonnegative stakes and an optional cash position. The usual solution is elementary, but its support-selection step is often presented as a Lagrange-multiplier exercise. This note records a shorter viewpoint. Cash is an implicit position in every outcome: holding cash fraction c is equivalent to already staking cq_i on outcome i, where q_i is the state price (equivalently, the inverse decimal odds). On any active support, the effective total stake must therefore equal the true probability vector, so the additional explicit stake is $p_i - cq_i$. This immediately yields the threshold rule $p_i/q_i > c$, the closed form for the optimal stake vector, and a one-pass greedy algorithm after sorting outcomes by p_i/q_i. The result is standard in substance, but the implicit-position interpretation gives a compact derivation and a convenient way to remember the formula.

1 Introduction

The Kelly criterion maximizes expected logarithmic growth and occupies a central place in gambling, portfolio theory, and information theory [4, 1, 8, 5]. In the horse-race model, or more generally in a finite state market with state-contingent claims, the optimization problem is concave and completely explicit [4, 2]. For a single multinomial event with mutually exclusive outcomes, explicit treatments also appear in the pari-mutuel literature, notably in Rosner's early analysis [6] and in the direct closed-form solution of Smoczynski and Tomkins [7]. The remaining practical question is therefore not whether the problem can be solved, but how to present the solution in a way that makes the active-set structure transparent and easy to remember.

This note gives a short answer. The key observation is that cash itself is a state-contingent position. If the bettor keeps cash fraction c, then in state i this contributes wealth c; equivalently, it is as if the bettor were already holding an implicit stake cq_i on outcome i. Once that is recognized, the fixed-support problem collapses to the classical full-investment Kelly rule, and the support-selection problem becomes a monotone threshold test. The resulting algorithm is greedy: sort by the edge ratio p_i/q_i and keep adding outcomes while that ratio exceeds the current cash level.

Nothing here changes the classical theory. The point is expository. The contribution is a particularly short state-price derivation: cash is an all-state claim, so active bets merely top up favorable outcomes from cq_i to p_i. This isolates the idea that makes the formulas memorable and makes the active-set structure almost self-evident.

2 Model

Consider a single event with outcomes $i = 1, \ldots, n$. Let

$$p_i > 0, \qquad \sum_{i=1}^{n} p_i = 1,$$

be the bettor's subjective probabilities. Outcome i is available at decimal odds $O_i > 1$, and we write

$$q_i := \frac{1}{O_i} > 0$$

for the corresponding state price or implied probability. The overround is $\sum_i q_i$, which may exceed 1.

A strategy consists of nonnegative stakes $x_i \geq 0$ and a cash holding $c \geq 0$ satisfying

$$c + \sum_{i=1}^{n} x_i = 1.$$

If outcome i occurs, terminal wealth is

$$W_i(c, x) = c + \frac{x_i}{q_i}.$$

The single-period full Kelly problem is

$$\max_{c \geq 0,\, x_i \geq 0} \; G(c, x) := \sum_{i=1}^{n} p_i \log\left(c + \frac{x_i}{q_i}\right) \quad \text{subject to} \quad c + \sum_{i=1}^{n} x_i = 1. \tag{1}$$

Since the feasible set is compact and the objective is strictly concave, the maximizer is unique.

It is convenient to define the *edge ratios*

$$r_i := \frac{p_i}{q_i}.$$

These are the bettor's probabilities divided by market-implied probabilities. They are also the terminal wealth levels that arise on active outcomes at the optimum.

3 A full-investment lemma

The following elementary lemma is the only optimization fact we need.

Lemma 1 (Full-investment Kelly rule)**.** *Let $a_i > 0$ with $\sum_{i=1}^{m} a_i = A$, and let $z_i > 0$ satisfy $\sum_{i=1}^{m} z_i = A$. Then*

$$\sum_{i=1}^{m} a_i \log z_i$$

is uniquely maximized at $z_i = a_i$ for all i.

Proof. Write

$$\sum_{i=1}^{m} a_i \log z_i - \sum_{i=1}^{m} a_i \log a_i = \sum_{i=1}^{m} a_i \log\left(\frac{z_i}{a_i}\right).$$

By concavity of log,

$$\sum_{i=1}^{m} \frac{a_i}{A} \log\left(\frac{z_i}{a_i}\right) \leq \log\left(\sum_{i=1}^{m} \frac{a_i}{A} \frac{z_i}{a_i}\right) = \log\left(\frac{1}{A} \sum_{i=1}^{m} z_i\right) = 0.$$

Equality holds only when z_i/a_i is constant in i, and the constraint $\sum_i z_i = \sum_i a_i$ forces that constant to be 1. □

In the betting language, if one is forced to invest the entire amount A across a collection of mutually exclusive outcomes, then the log-optimal total stakes are proportional to the probabilities themselves.

4 Fixed-support optimization

Let $A \subseteq \{1, \dots, n\}$ be a candidate active set. Write

$$P_A := \sum_{i \in A} p_i, \qquad Q_A := \sum_{i \in A} q_i.$$

We ask for the best strategy whose positive stakes are confined to A.

Proposition 2 (Optimal strategy on a fixed support)**.** *Let $A \subseteq \{1, \dots, n\}$ be nonempty.*

(a) *If $A \neq \{1, \dots, n\}$ and $Q_A < 1$, define*

$$c_A := \frac{1 - P_A}{1 - Q_A}. \tag{2}$$

If $p_i > c_A q_i$ for every $i \in A$, then the unique maximizer of (1) among strategies with $x_j = 0$ for $j \notin A$ is

$$x_i^{(A)} = p_i - c_A q_i \quad (i \in A), \qquad x_j^{(A)} = 0 \quad (j \notin A), \tag{3}$$

with cash c_A. The resulting terminal wealth is

$$W_i = \frac{p_i}{q_i} = r_i \quad (i \in A), \qquad W_j = c_A \quad (j \notin A). \tag{4}$$

(b) *If $A = \{1, \dots, n\}$, then the unique support-constrained maximizer is $c_A = 0$ and $x_i^{(A)} = p_i$ for all i.*

Proof. For case (a), Fix a strategy with support contained in A. For $i \in A$, define the *effective total stake*

$$y_i := x_i + c q_i.$$

This is natural because cq_i staked on outcome i would pay exactly c in state i; cash is therefore an implicit position in every outcome. For $i \in A$ we have

$$W_i = c + \frac{x_i}{q_i} = \frac{y_i}{q_i},$$

while for $j \notin A$ we have $W_j = c$.

Now impose the budget constraint with support A:

$$1 = c + \sum_{i \in A} x_i.$$

Using $y_i = x_i + cq_i$ and the definition (2) of c_A,

$$\sum_{i \in A} y_i = \sum_{i \in A} x_i + c_A Q_A = 1 - c_A + c_A Q_A = P_A.$$

Hence, for strategies on support A, maximizing expected log wealth is equivalent to maximizing

$$\sum_{i \in A} p_i \log\left(\frac{y_i}{q_i}\right) + (1 - P_A) \log c_A$$

over $y_i > 0$ with $\sum_{i \in A} y_i = P_A$. The terms involving q_i and c_A are constant, so Lemma 1 gives the unique maximizer $y_i = p_i$ for $i \in A$. Therefore $x_i = p_i - c_A q_i$ on A, and (4) follows immediately.

For case (b), cash is identically zero and the problem reduces directly to the full-investment lemma with total stake 1, so $x_i = p_i$. □

The proposition explains the formula. If cash level c_A is held back, then the bettor already has an implicit stake $c_A q_i$ in every outcome i. On the active support, the total stake should therefore be p_i, so the additional explicit stake is $p_i - c_A q_i$.

The positivity condition in Proposition 2 can be written as

$$\frac{p_i}{q_i} > c_A. \tag{5}$$

Thus an outcome is active exactly when its edge ratio exceeds the prevailing cash level.

5 Greedy support selection

Sort the outcomes so that

$$r_1 \geq r_2 \geq \cdots \geq r_n.$$

For $k = 0, 1, \ldots, n$, let

$$A_k := \{1, \ldots, k\}, \qquad P_k := \sum_{i=1}^{k} p_i, \qquad Q_k := \sum_{i=1}^{k} q_i,$$

with the convention $P_0 = Q_0 = 0$ and $c_0 = 1$. Whenever $Q_k < 1$, define

$$c_k := \frac{1 - P_k}{1 - Q_k}.$$

If $k = n$, the formula reduces to $c_n = 0$.

Theorem 3 (Greedy characterization of the Kelly support)**.** *Start from $A_0 = \varnothing$ and cash level $c_0 = 1$. At step k, if $k = n$ stop. Otherwise compare the next ratio r_{k+1} with c_k.*

- *If $r_{k+1} > c_k$, add outcome $k+1$ to the support and update cash to*

$$c_{k+1} = \frac{1 - P_{k+1}}{1 - Q_{k+1}}.$$

- *If $r_{k+1} \leq c_k$, stop.*

Let k^* be the stopping index. Then the unique maximizer of (1) is

$$c^* = c_{k^*}, \qquad x_i^* = \begin{cases} p_i - c^* q_i, & 1 \leq i \leq k^*, \\ 0, & i > k^*, \end{cases} \tag{6}$$

and the optimal terminal wealth is

$$W_i^* = \max\left\{c^*, \frac{p_i}{q_i}\right\}. \tag{7}$$

In particular, the active set is a prefix after sorting by p_i/q_i.

Proof. By Proposition 2, the optimizer on A_k is obtained by setting cash equal to c_k and then taking stakes $x_i = p_i - c_k q_i$ on A_k.

We first show the monotonic update rule. A short calculation gives

$$c_{k+1} < c_k \iff \frac{1 - P_{k+1}}{1 - Q_{k+1}} < \frac{1 - P_k}{1 - Q_k} \iff p_{k+1} > c_k q_{k+1} \iff r_{k+1} > c_k.$$

So adding an outcome strictly decreases cash exactly when that outcome's edge ratio exceeds the current cash level. Because the r_i are sorted in decreasing order, once $r_{k+1} \le c_k$ holds, it also holds for every later outcome. Therefore the active set can only grow as a prefix.

It remains to justify the stopping rule. At the support optimizer for A_k, moving an infinitesimal amount $\varepsilon > 0$ from cash to a previously inactive outcome $j > k$ changes the objective by

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0^+} G = -\frac{\partial G}{\partial c} + \frac{\partial G}{\partial x_j}.$$

At the optimizer on A_k, Proposition 2 yields wealth levels $W_i = r_i$ for $i \le k$ and $W_i = c_k$ for $i > k$. Hence

$$\frac{\partial G}{\partial x_j} = \frac{p_j}{q_j W_j} = \frac{p_j}{q_j c_k} = \frac{r_j}{c_k}.$$

Also

$$\frac{\partial G}{\partial c} = \sum_{i=1}^{n} \frac{p_i}{W_i} = \sum_{i \le k} q_i + \sum_{i > k} \frac{p_i}{c_k} = Q_k + \frac{1 - P_k}{c_k} = 1,$$

because $c_k(1 - Q_k) = 1 - P_k$. Therefore the entry derivative is

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0^+} G = \frac{r_j}{c_k} - 1. \tag{8}$$

So outcome j is worth adding if and only if $r_j > c_k$.

If $r_{k+1} > c_k$, then the next outcome has positive entry derivative and the support A_k cannot be globally optimal. If $r_{k+1} \le c_k$, then every inactive outcome has nonpositive entry derivative, and Proposition 2 already gives optimality with respect to all changes inside the current support. Since the feasible set is a simplex and G is concave, this proves that the strategy on A_k is globally optimal. The formula (7) is just a restatement of the active and inactive wealth levels. □

Remark 4 (Binary reduction)**.** *For a binary event, if only outcome* 1 *is active then*

$$c^* = \frac{1 - p_1}{1 - q_1}, \qquad x_1^* = p_1 - c^* q_1 = \frac{p_1 - q_1}{1 - q_1},$$

which is the classical one-bet Kelly fraction written in state-price form.

Remark 5 (Interpretation)**.** *The optimal terminal wealth has a simple clipped form: on outcomes with sufficiently large edge ratio p_i/q_i, the bettor raises wealth to the level p_i/q_i; on the rest, wealth is left at the cash floor c^*. The quantity c^* is therefore the cutoff separating active from inactive states.*

6 Concluding comment

For a single multinomial event, the full Kelly solution is completely explicit; see in particular [6, 7] for earlier formulations in the pari-mutuel setting. What is useful in practice is not another closed form, but a way to derive it without losing the intuition. The implicit-position viewpoint does exactly that: cash is an all-state claim, so active bets only need to top up each favorable outcome from cq_i to p_i. Once written this way, the threshold rule and greedy support selection are immediate.

References

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